

# Fernando Fernández Andueza

## Gameplay and AI Programmer

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Valencia, Spain. EU citizen, open to relocation within Spain and to remote work. Spanish (native).

### Profile

I built all four enemy archetypes for Beerserker: The Stolen Brew in StateTree, with C++ and Blueprints, and wrote XEMA, my own C++ and OpenGL engine, with an ECS and deferred PBR rendering. I work with Unreal and Unity, and before games I spent two and a half years as a fullstack developer on ERP software.

### Technical skills

|                               |                                                                                                                                                 |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Engines &amp; AI</b>       | Unreal Engine 5 (C++ & Blueprints), Unity (C#), StateTree, Behaviour Trees, EQS, AI Perception, Animation systems, Object pooling               |
| <b>Languages</b>              | C++, C#, C, ARM64 assembly, Lua                                                                                                                 |
| <b>Graphics &amp; systems</b> | OpenGL 4.5 DSA, Deferred shading / PBR, Vertex & fragment shaders, ECS architecture, Multithreading & job systems, Memory management & pointers |
| <b>Tools</b>                  | Visual Studio debugger, Unreal Insights, RenderDoc, Git, Perforce, Plastic SCM                                                                  |
| <b>Outside games</b>          | TypeScript / JavaScript, HTML / CSS, Vue.js, SQL Server, Java, VB.NET                                                                           |

### Experience

#### Gameplay and AI Programmer, Beerserker: The Stolen Brew

Sep 2025 – Jul 2026

ESAT final-year project, team of about 20. Developed by Grumpy Games and ESAT, published on Steam by ESAT (Early Access, free).

- Implemented the four enemy archetypes (melee, ranged, explosive and a tank mini-boss) with StateTree in C++ and Blueprints, with custom tasks and conditions.
- Built an object pool as a plugin, reusable in other projects, for the many enemies, particles, decals and projectiles.
- Used EQS to pick where the ranged enemy retreats to, based on distance to the player and line of sight. The tank favours its less-used melee attacks, so it doesn't keep repeating the same one.
- Built the enemy spawners, set up and uploaded the Steam builds, and took part in designing the architecture of a custom GAS-based action system.

#### Gameplay Programmer, Roadtrip: The Engine of Madness (Oniric Tales, Unity)

Dec 2024 – Jun 2025

- Designed and built core gameplay systems: a component-based dialogue system, an AI perception system integrated with the behaviour trees, and a radio communication system.
- Also worked on the Behavior Graph, general gameplay mechanics, profiling and bug fixing.

#### Fullstack Developer, AHORA

Sep 2020 – Mar 2023

- Enterprise management system: frontend with Vue.js, TypeScript and DevExtreme; backend in VB and C# on SQL Server.
- Built products, components and extensions with FlexyGo, an in-house low-code framework.
- Handled client incidents, often on site. Worked in Scrum.

### Selected projects

#### XEMA (custom C++ engine)

C++, OpenGL 4.5

- ECS, deferred PBR, shadow mapping, instancing, a JobSystem for async mesh loading, Lua scripting and PhysX.
- Ported to Nintendo Switch, tested only on an emulator. PS4 port in progress on a devkit: the ECS and simple-geometry rendering work; mesh loading, textures, instancing, lights, shadows and PBR are still missing.

#### Abandoned Hospital

Unreal Engine 5, C++

- Class exercise with AI Perception and EQS on a behaviour tree; I later rebuilt it on my own, with the AI rewritten in StateTree.

### Education & certifications

#### HND in Video Game Programming, ESAT (Escuela Superior de Arte y Tecnología), Valencia

2023–2026

Specialising in C++: my own engine from scratch, procedural generation (Wave Function Collapse, cellular automata, L-systems), a templated FSM, A\* pathfinding, and assembly to find and fix bottlenecks in inner loops.

#### DAM (Multiplatform Application Development), IES El Grao, Valencia

2018–2020

Java for desktop and Android, relational databases, HTML and CSS, OOP fundamentals and version control.

#### Certified Scrum Master & Product Owner, Scrum.org / International Agile Institutes

Professional credentials